CoE toy universe: geometry, Migkas fits and the observable list

Letter for Chris, 27 October 2026. Plain numbers from files on the working machine; no pass/fail calls.

1 The geometry

The model has one preferred direction: the line from us to the centre of expansion (CoE), at distance D. Our observable universe is a sphere of radius R around us (Fig. 1).

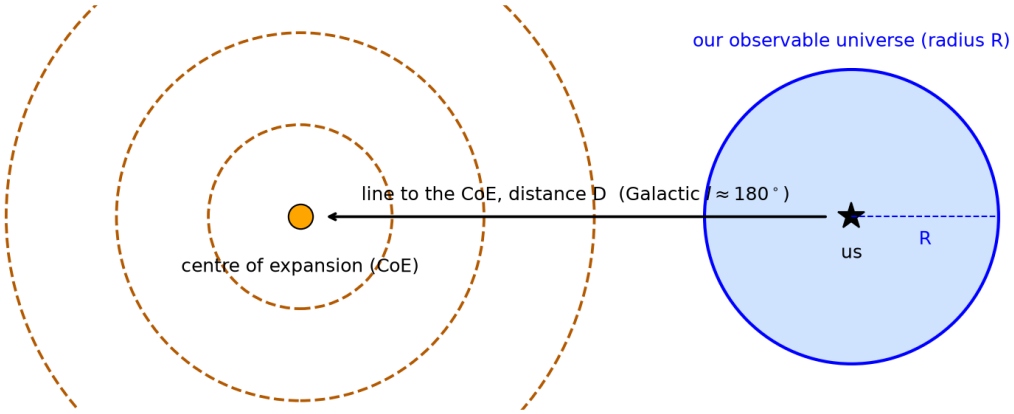


Figure 1: Schematic: CoE, our position at distance D, our observable universe around us. Script `geometry_schematic.py`.

Everything else is free to roll about the line to the CoE. The orange arc in Fig. 2 is the set of points as far from the CoE as we are, so they sit at the same CoE potential. Where it crosses the sky it gives the critical points P1, P2 and their antipodes.

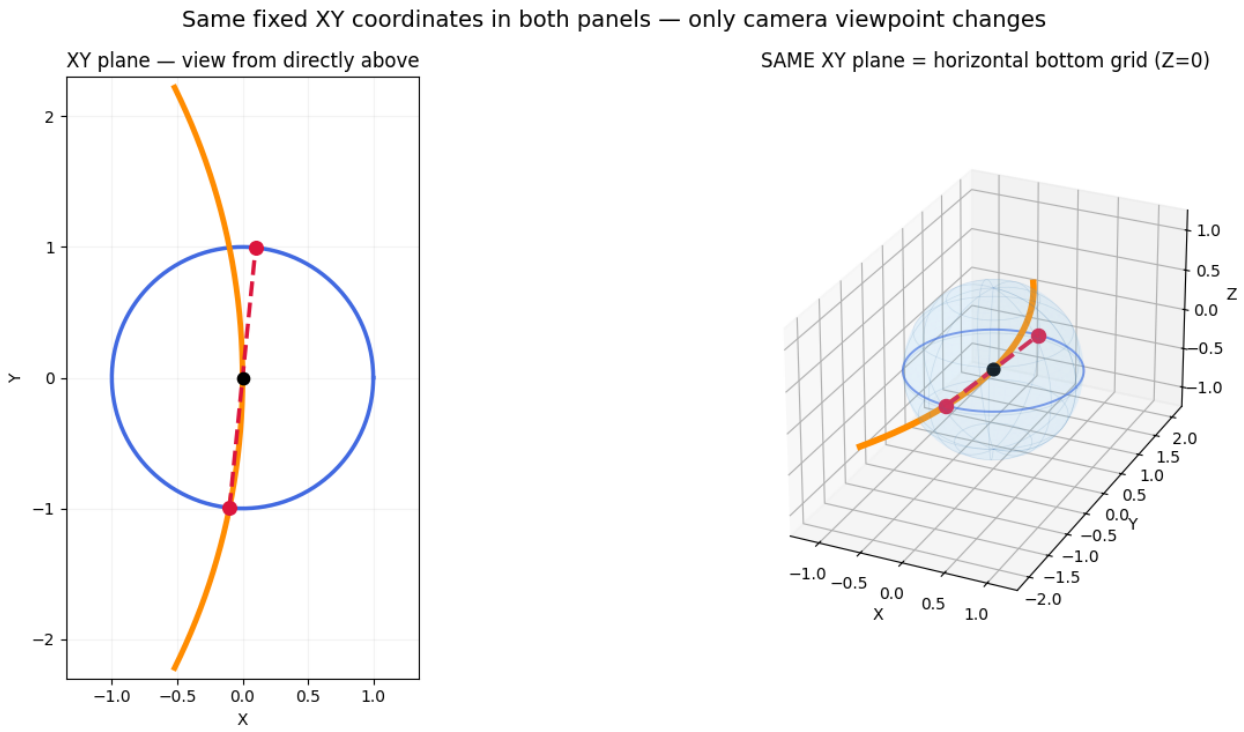


Figure 2: Equal-distance arc and critical points P1, P2 (figure by C. Alameddin).

2 Migkas $A/A_{\rm all}$ and $B/B_{\rm all}$ vs angle

Migkas et al. (2020, A&A 636, A15) give the L_X–T normalisation A and slope B in sliding 90° cones; digitised at 71 points. Adjacent cones share most clusters, so χ^2 is descriptive.

What A and B are. Migkas fit the cluster X-ray luminosity–temperature scaling relation in each patch of sky, $L_X/E(z) = A\,(T/4\,\mathrm{keV})^B$ (with L_X in units of $10^{44}\,\mathrm{erg/s}$). A is the normalisation: how bright a cluster of given temperature appears, which depends on the distance inferred from its redshift, so it tracks the local expansion rate ($A \propto H_0^{-2}$ at fixed physics). B is the slope: how luminosity scales with temperature, which depends on the cluster sample and foreground absorption rather than on distance. $A_{\rm all}$ and $B_{\rm all}$ are the full-sky values, so $A/A_{\rm all} = 1$ means that direction looks like the sky average; values above or below 1 show directional anisotropy.

A and B together (Fig. 3; source `migkas_bars_rotation_summary.json`): sheet shell model with a one-cycle phase rotation, Amp $= \mp 0.06$, $\varphi = 0.35\,\mathrm{rad}$. R^2 vs digitised points: A 0.446 → 0.712, B 0.483 → 0.785. Best shared $\varphi = 0.522\,\mathrm{rad}$: A 0.730, B 0.776. Against the sheet series: A 0.501 → 0.739, B 0.372 → 0.733.

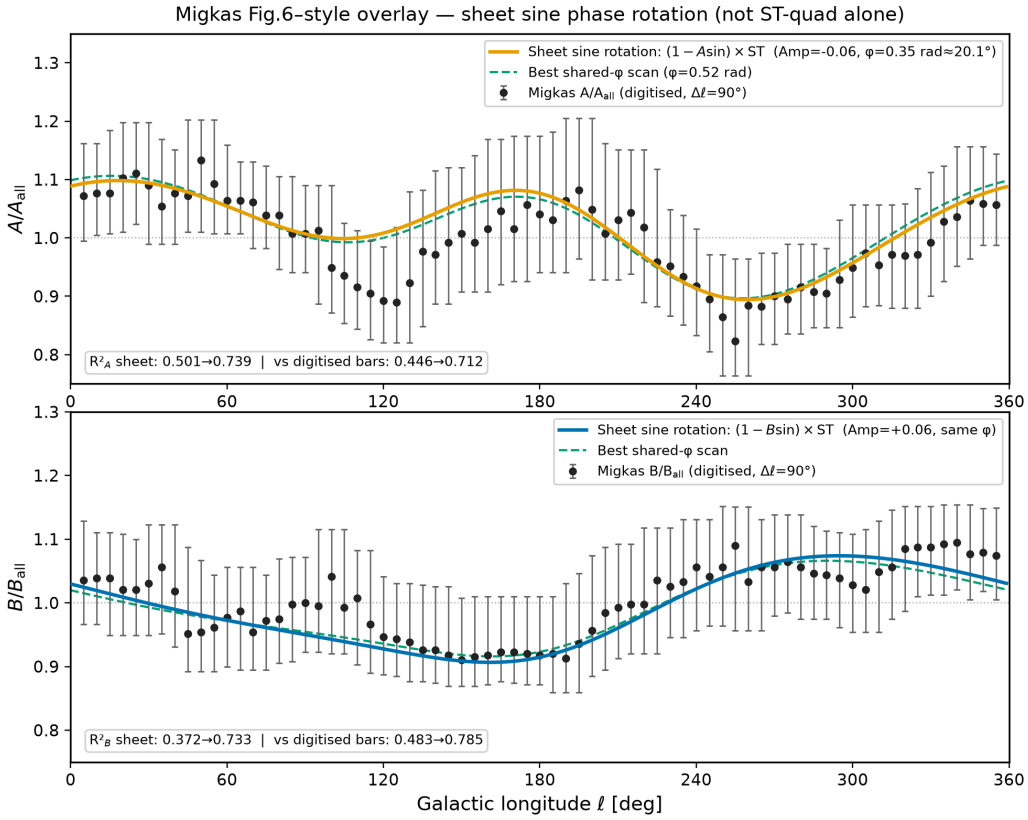


Figure 3: $A/A_{\rm all}$ (top) and $B/B_{\rm all}$ (bottom) vs longitude with the rotated shell model. Script `R04_migkas_rotation.py`.

3 Placing the model on the sky

The geometry fixes only the CoE line, so to compare with sky maps we must orient it: point the CoE axis at the sky and choose a roll about that axis. This section shows that orientation step and how the predicted bands and residual spots then fall on the Migkas maps.

Placing the model needs the CoE direction (near $l \approx 180°$) and one roll about the CoE axis; expected roll band 45–80°, 60° central. Fig. 5 shows the orientation sequence and the Migkas 2D map overlays after rotation. Note on roll: Figs. 5 and 6 were made with a roll of about 80° (80.8°), and Fig. 7 with a roll of 45°. These are illustrative orientations; later in the letter (Results from today) the roll is pinned using the axis of evil together with the other fits.

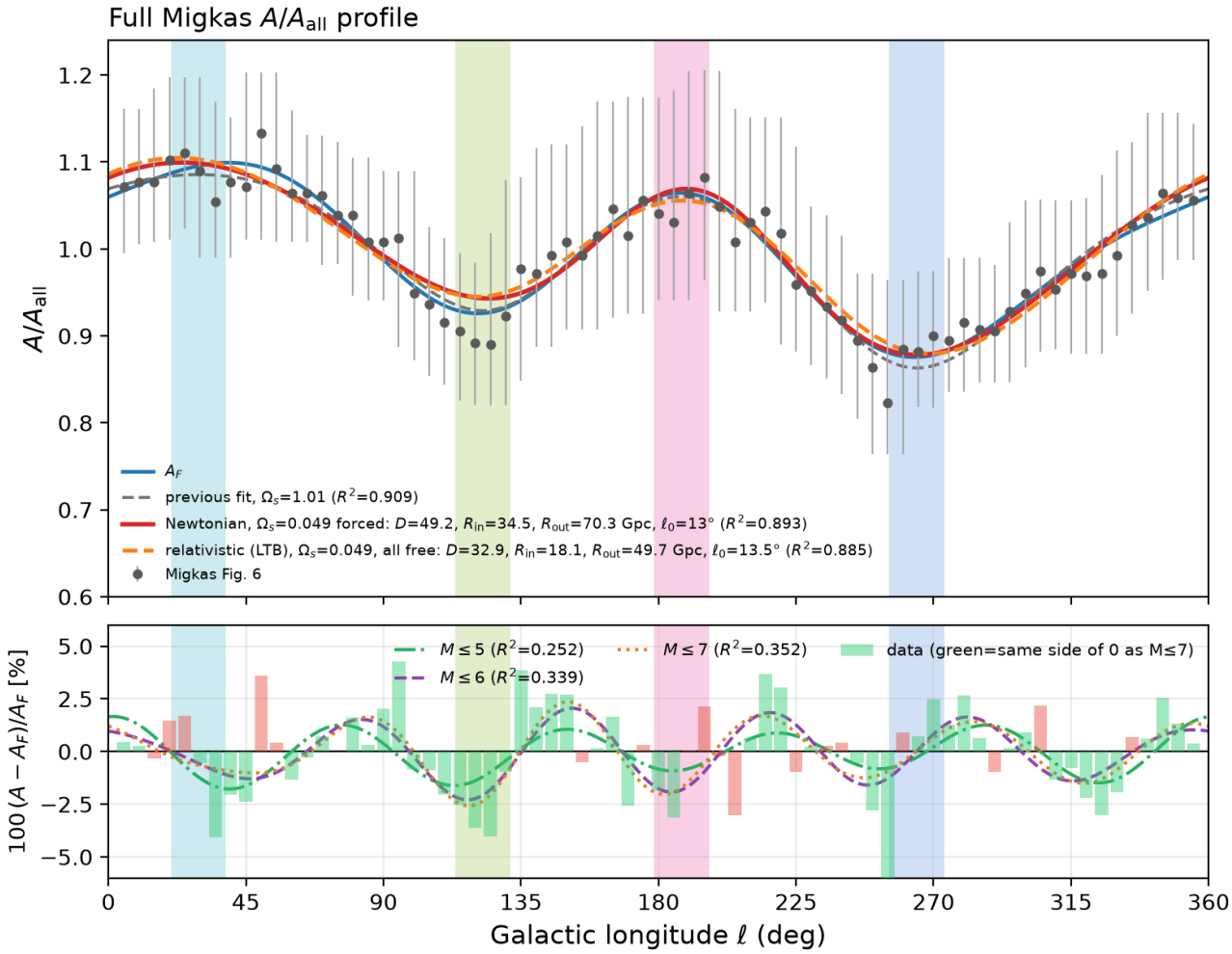


Figure 4: Full Migkas $A/A_{\rm all}$ profile vs Galactic longitude with the four MAX/MIN bands shaded; lower panel: residual bars $100(A - A_F)/A_F$ with $M \le 5, 6, 7$ harmonic curves. Script `Fig04_amp_full_stripes.py` (from `14_make_fig04_amp_full.py`).

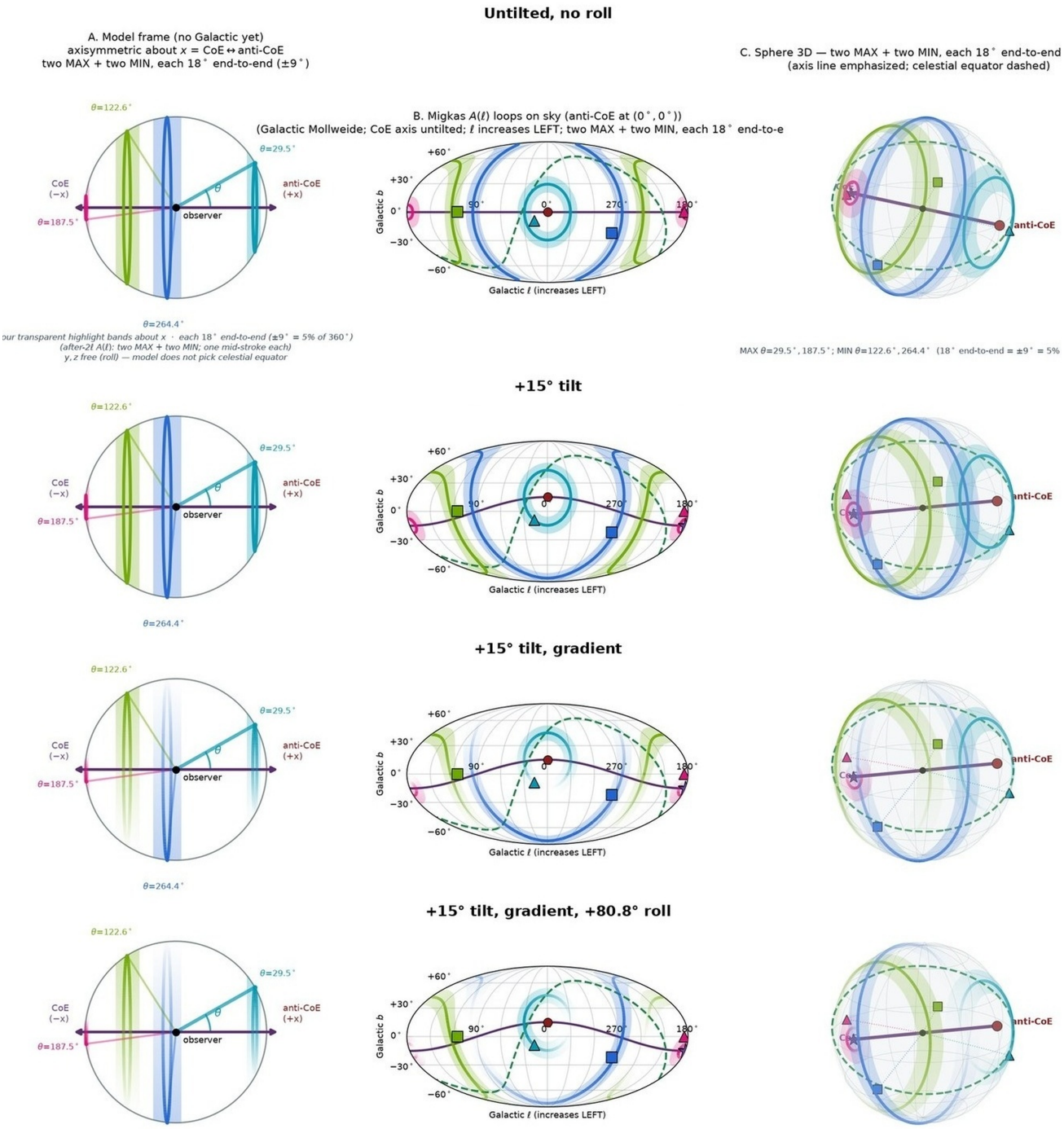


Figure 5: Sky-orientation sequence (untilted, +15° tilt, radial gradient, +80.8° roll about the CoE axis). Maker `make_fig05_sky_orientation.py`; split by `split_fig8.py`.

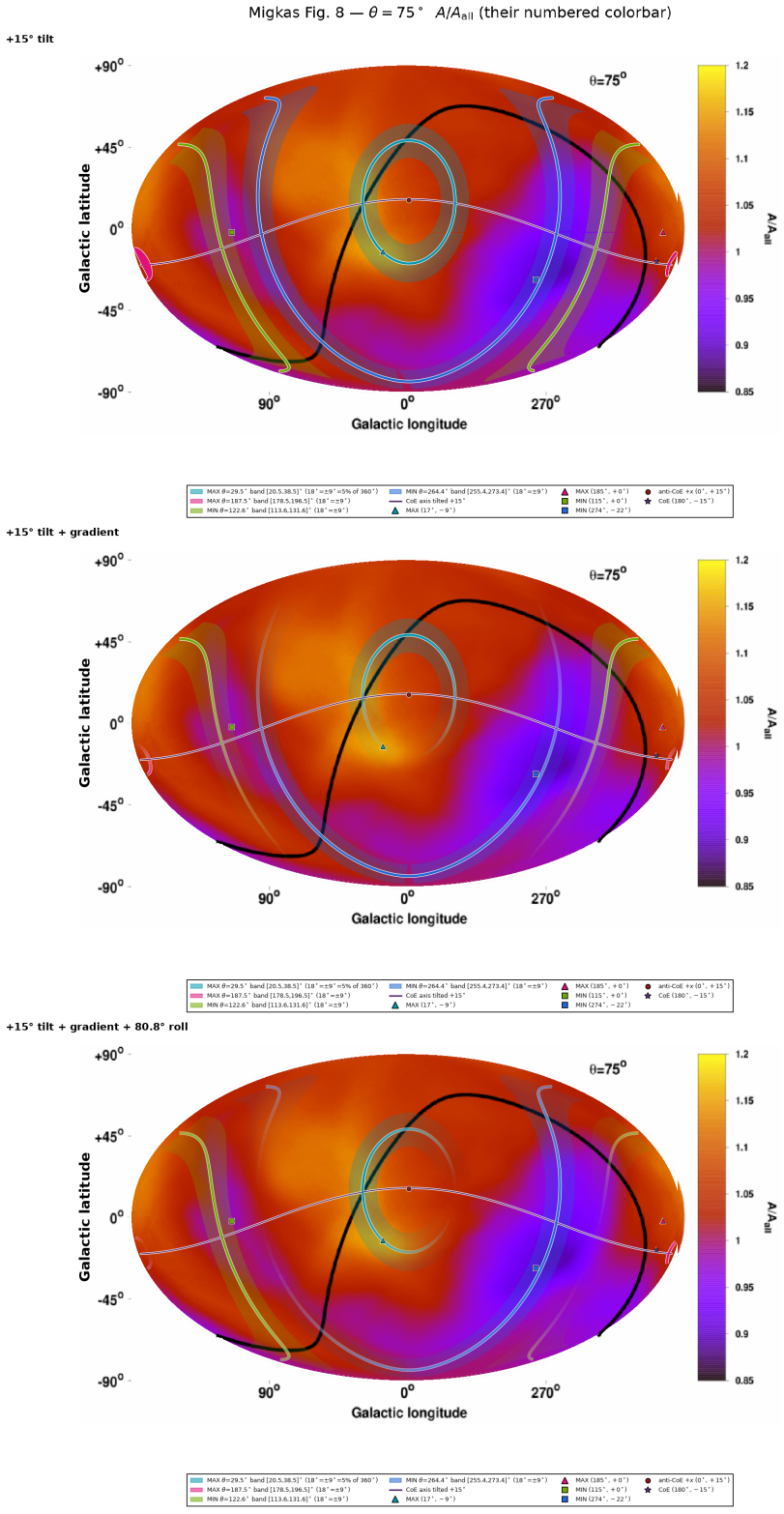


Figure 6: Predicted MAX/MIN bands (two of each, 18° wide) on the Migkas Fig. 8 2D map. Maker `make_migkas_fig8_overlay_matched.py`; split by `split_fig8.py`.

The A and B residuals (after removing shared $n = 1$–3 harmonics, 25° running mean) are put on the sky with the CoE at $b = -15°$ and roll 45° (Fig. 7).

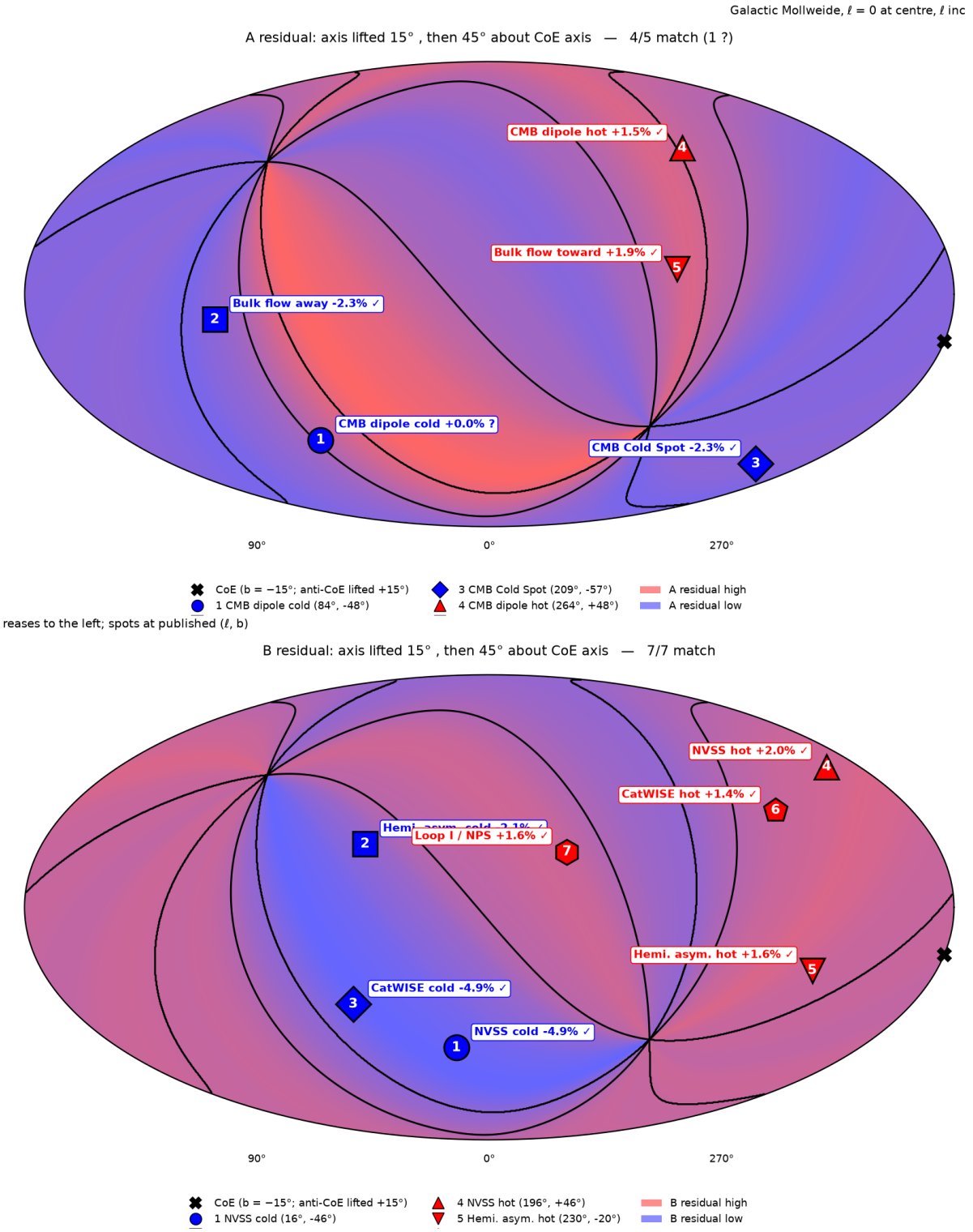


Figure 7: A (top, spots 1–5) and B (bottom, spots 6–12) residual maps, CoE $b = -15°$, roll 45°. Script `chris_AB_residual_spots.py` (from `23_sky_mapA_al.py`).

4 Where the dipoles land

Several independent large-scale anomalies (CMB dipole, bulk flow, radio and quasar number-count dipoles, hemispherical asymmetry, Cold Spot, axis of evil) each point in a direction on the sky. This map puts them all, with their antipodes, on one Galactic frame alongside the model CoE direction so their positions relative to the geometry can be read directly.

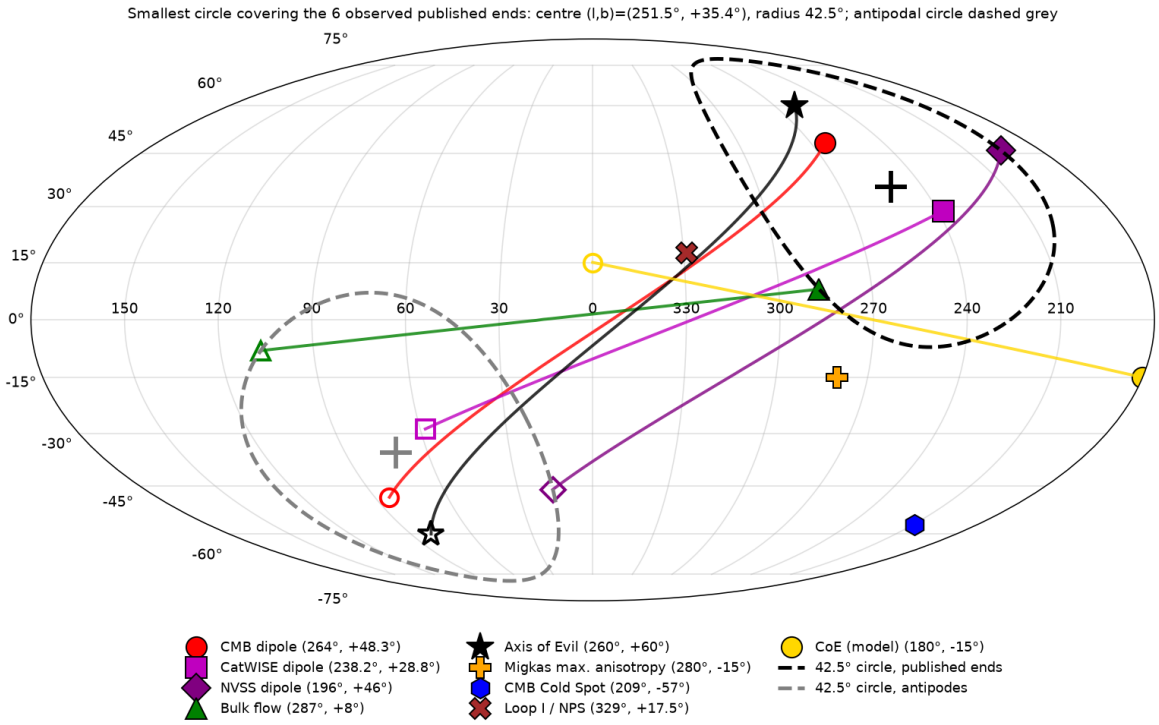


Figure 8: Published dipole and anomaly directions (filled) and antipodes (open), with the model CoE direction $(180°, -15°)$. Migkas maximum anisotropy $(280°, -15°)$ from Migkas et al. (2021). Script `dipoles_sky_map.py`.

5 The observables from the CoE point of view

Here each published direction from the previous map is read against the A or B residual maps: if the geometry is right, a hot pole should land on a positive residual and a cold pole on a negative one. Each spot is tied to A (normalisation: distance, expansion, velocity) or B (slope: sample composition, foreground) from the physics before reading any sign. A hot spot needs residual > 0, a cold spot < 0. Bootstrap 1σ per spot is 3–6 points.

Observable	(l, b)	tie	residual [%]	sign
CMB dipole cold	$(84, -48)$	A	$+0.024$	on zero line
CMB dipole hot	$(264, +48)$	A	$+1.538$	matches
Bulk flow away	$(107, -8)$	A	-2.272	matches
Bulk flow toward	$(287, +8)$	A	$+1.903$	matches
CMB Cold Spot	$(209, -57)$	A	-2.331	matches
NVSS cold	$(16, -46)$	B	-4.872	matches
NVSS hot	$(196, +46)$	B	$+2.028$	matches
Hemispherical asym. cold	$(50, +20)$	B	-2.110	matches
Hemispherical asym. hot	$(230, -20)$	B	$+1.621$	matches
CatWISE cold	$(58, -31)$	B	-4.873	matches
CatWISE hot	$(238, +31)$	B	$+1.388$	matches
Loop I / NPS	$(329, +17.5)$	B	$+1.598$	matches
Score: A 4/5, B 7/7, total 11/12 (source `lee_results.json`, `R26_29_sky_chain.py`)				

CMB dipole (A; our motion → distance/velocity). Planck 2018 I (A&A 641, A1): 3.362 mK toward $(264.0°, +48.3°)$, i.e. 369.8 km/s. Hot pole +1.54, cold pole +0.02 (on the zero line, marked "?").

Bulk flow (A; recession velocity → distance). Watkins, Feldman & Hudson 2009 (MNRAS 392, 743): 407 ± 81 km/s toward $(287 \pm 9°, 8 \pm 6°)$. Toward +1.90, away −2.27.

CMB Cold Spot (A; Eridanus supervoid changes expansion along that sightline). Vielva et al. 2004 (ApJ 609, 22); Cruz et al. 2005 (MNRAS 356, 29): $\sim -70\,\mu$K at $(209°, -57°)$. Supervoid: Szapudi et al. 2015 (MNRAS 450, 288). Residual −2.33.

NVSS radio dipole (B; source density → sample). Singal 2011 (ApJL 742, L23); Rubart & Schwarz 2013 (A&A 555, A117): amplitude several times the kinematic expectation, direction near the CMB dipole. Cold −4.87, hot +2.03.

Hemispherical power asymmetry (B; power direction → sample). Eriksen et al. 2004 (ApJ 605, 14); Planck 2018 VII (A&A 641, A7): dipolar modulation $A \approx 0.07$ at low ℓ. Cold −2.11, hot +1.62.

CatWISE quasar dipole (B; matter dipole → sample). Secrest et al. 2021 (ApJL 908, L51): $\mathcal{D} = 0.01554$ toward $(238.2°, +28.8°)$, about twice the kinematic value, 4.9σ. Cold −4.87, hot +1.39.

Loop I / North Polar Spur (B; soft X-ray foreground → slope via $N_{\rm H}$ and soft background). Berkhuijsen, Haslam & Salter 1971 (A&A 14, 252); centre $\approx (329°, +17.5°)$. Residual +1.60.

Axis of Evil. Land & Magueijo 2005 (PRL 95, 071301): quadrupole–octopole axis $\approx (260°, +60°)$. In the geometry it lies at a critical point of the equal-distance arc (Sec. 1, Sec. 4).

Preferred plane. Rakić & Schwarz 2007 (PRD 75, 103002; astro-ph/0703266): "The data require a preferred plane, whereupon the axis is just the normal direction. Rotational symmetry within that plane is ruled out at high confidence." The CoE geometry breaks symmetry about the CoE axis once the roll is set; the CoE axis is 81.9° from the Axis of Evil, i.e. about 8° from that plane.

6 Results from today

Three checks of how special the orientation is: how often random orientations score as well (look-elsewhere), how close a critical point lands to the axis of evil as the CoE direction and roll vary, and how planar the $\ell = 2, 3$ sky is about the axis of evil.

Look-elsewhere test (`lee_results.json`; 200,000 random orientations, CoE direction and roll free,

spots fixed): $p(\geq 11/12) = 0.0164$; paired (antipodes once) 6/7: $p = 0.0179$; joint with an AoE critical point at least as close: $p \approx 0.003$ (0.00295 for phase4b $D/R = 1.258$); roll only, CoE fixed: $p = 0.543$. At the 11/12 orientation the nearest critical point is 21.2° from the AoE (phase4b D/R).

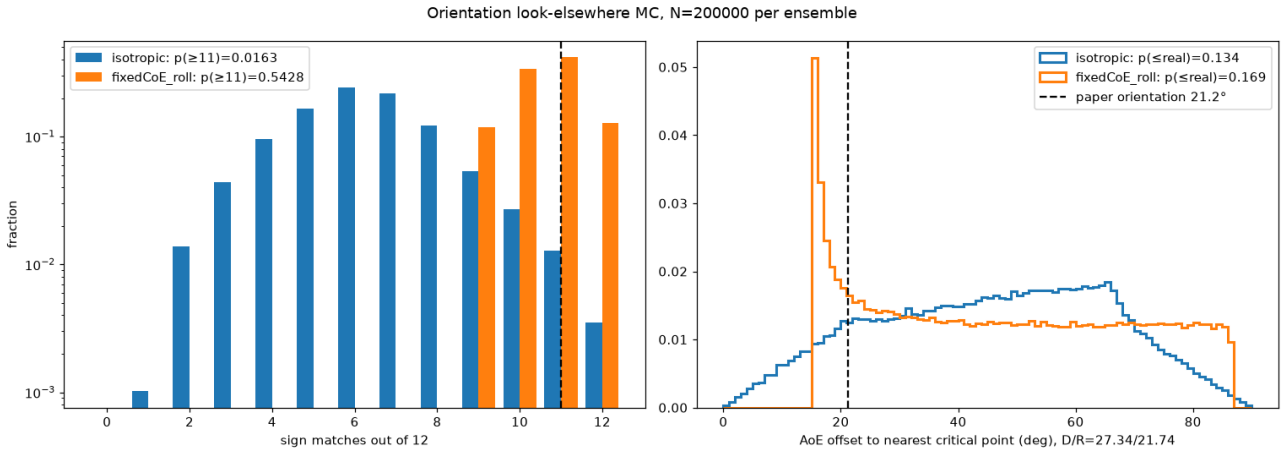


Figure 9: Score distributions over random orientations. Script `lee_hist.py`.

Orientation scan (`fig4_values.json`), nearest critical point to the AoE at roll 60° (expected band 45–80°):

CoE direction	$(180, 0)$	$(180, -15)$	$(180, +15)$
$D/R = 5$ (figure)	0.85°	2.33°	12.25°
$D/R = 1.258$ (phase4b)	18.45°	15.36°	5.60°

Best CoE direction at roll 60: 0.11° at $(166, +9)$ for $D/R = 5$; 0.35° at $(194, +13)$ for phase4b. Roll 300° gives 57–61°.

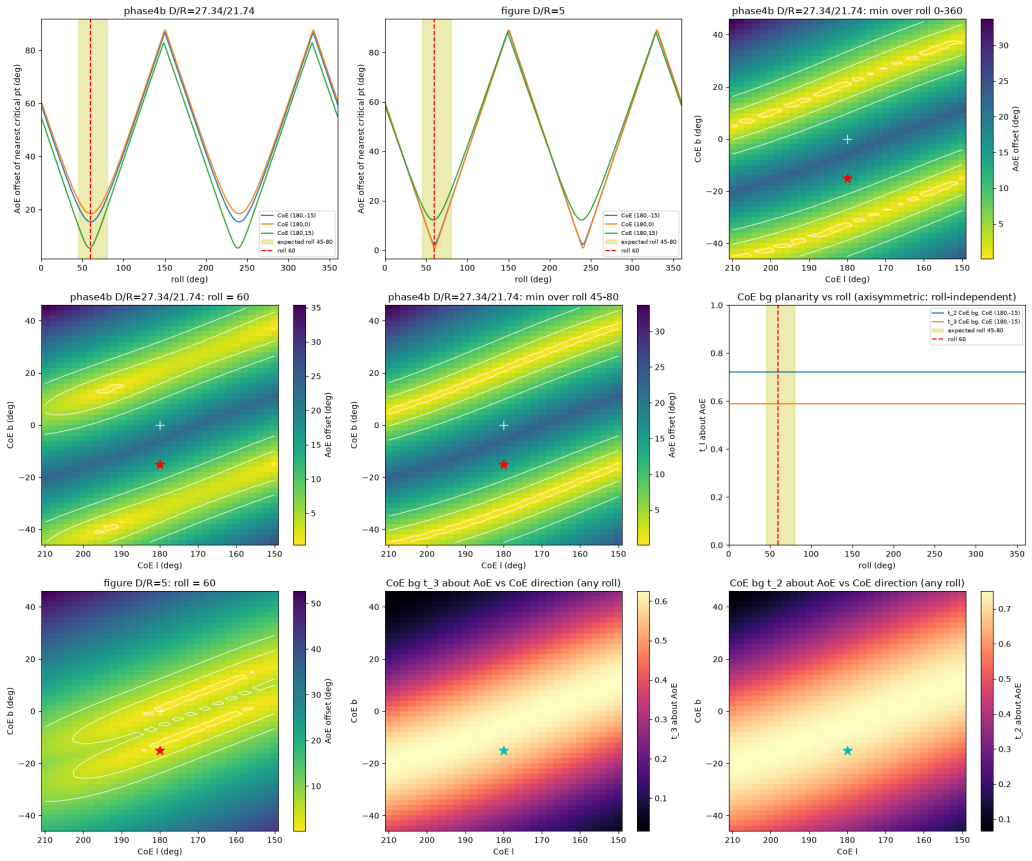


Figure 10: AoE offset vs CoE direction and roll; 45–80° band shaded. Script `fig4_orient_scan.py`.

Planarity about the AoE (`fig23_values.json`). The model sky was fitted to SMICA $\ell = 2$–130, so its planarity is inherited from the fit. The CoE background is axisymmetric about (180°, −15°), 81.9° from the AoE.

Sky	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
Planck SMICA	0.947	0.896	0.048	0.055
Fitted model (v3 seed 1)	0.934	0.903	0.046	0.060
CoE background alone	0.721	0.589	0.505	0.446
Random start	0.777	0.525	0.741	0.242

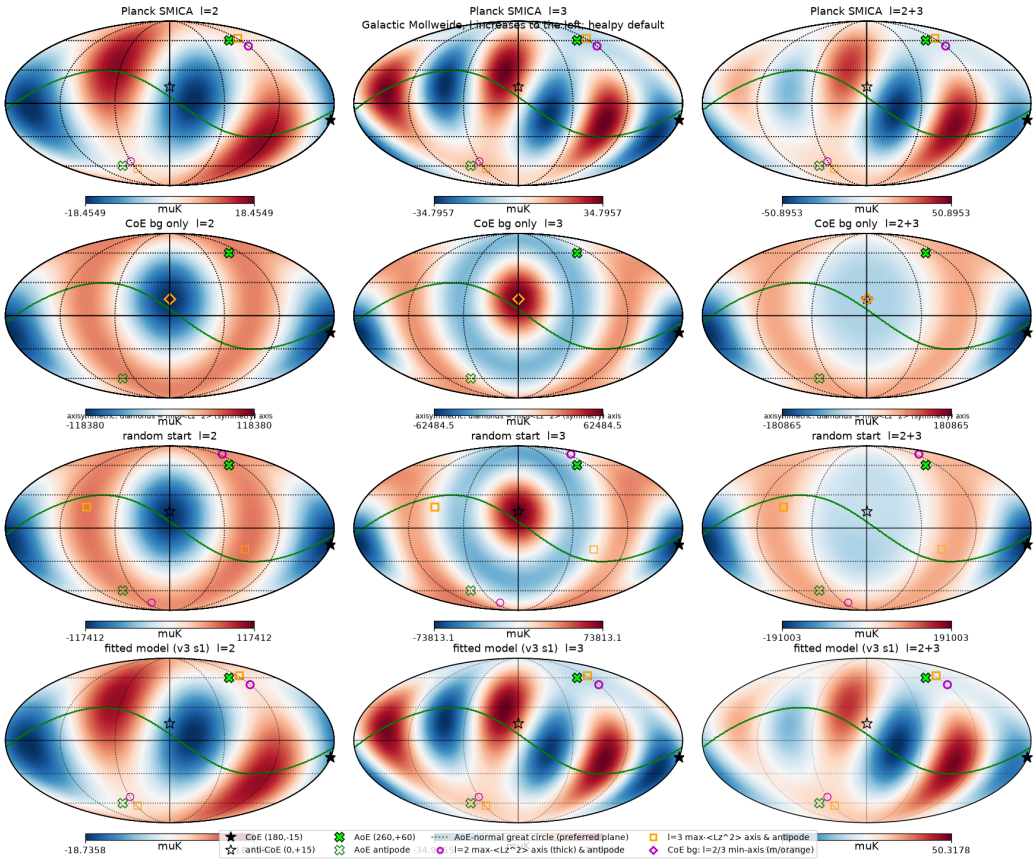


Figure 11: $\ell = 2, 3$ maps, same frame, CoE axis, AoE and plane marked. Script `fig1_maps.py`.

7 Next step

This is the planned next step; no results for it exist yet.

We propose an inverse problem that probes the plasma before recombination and the diffusive structure that came after it: the acoustic oscillations of the photon–baryon fluid and their Silk diffusion damping. The unknowns would be the pre-recombination source in the CoE geometry (the initial perturbation field and the sound horizon and damping scale it implies). The data would be the measured TT spectrum and the peak and damping-tail positions. The solver would fit these directly, instead of using the present phenomenological source spectrum.

Alongside it we will derive an exact analytical solution for the same problem: the tight-coupling acoustic wave equation with diffusion damping, solved in closed form on the CoE background. The numerical inversion can then be checked term by term against that solution.

Appendix: residual power vs Planck

This appendix compares the angular structure of the Migkas $A/A_{\rm all}$ curve with the CMB. The curve is decomposed into ring harmonics in Galactic longitude (order m), and its power per harmonic is set beside the Planck temperature power per multipole ℓ. These are different quantities, so the comparison is of shape only.

$A/A_{\rm all}$, locked fit, residuals and their Fourier modes (Fig. 12; computed by `A_fit_residual_fourier.py`, values in `A_fit_residual_fourier.json`). Locked CoE η-edge fit: $\chi^2 = 9.278$ (71 cones, mean of asymmetric errors), $r^2 = 0.745$ ($1 - \mathrm{SS_{res}}/\mathrm{SS_{tot}}$; the paper quotes 0.748 with its own definition). Least-squares harmonics of the residuals, $m = 1$–8, mean offset -0.0032. Largest modes: $m = 1$ amplitude 0.0316 (50.2% of residual harmonic power, peak at $l = 125°$); $m = 3$ 0.0219 (24.2%, first peak $l = 59°$); $m = 5$ 0.0151 (11.4%); $m = 2$ 0.0130 (8.5%). Modes 4, 6, 7, 8 together 5.8%. For reference (`N2_migkas_cov.json`): profile refit at $D = 34.96\,\mathrm{Gpc}$ $\chi^2 = 9.250$; isotropy 45.14; with overlap covariance CoE 3.88, isotropy 8.99, free dipole 6.90, dipole+quadrupole 5.20.

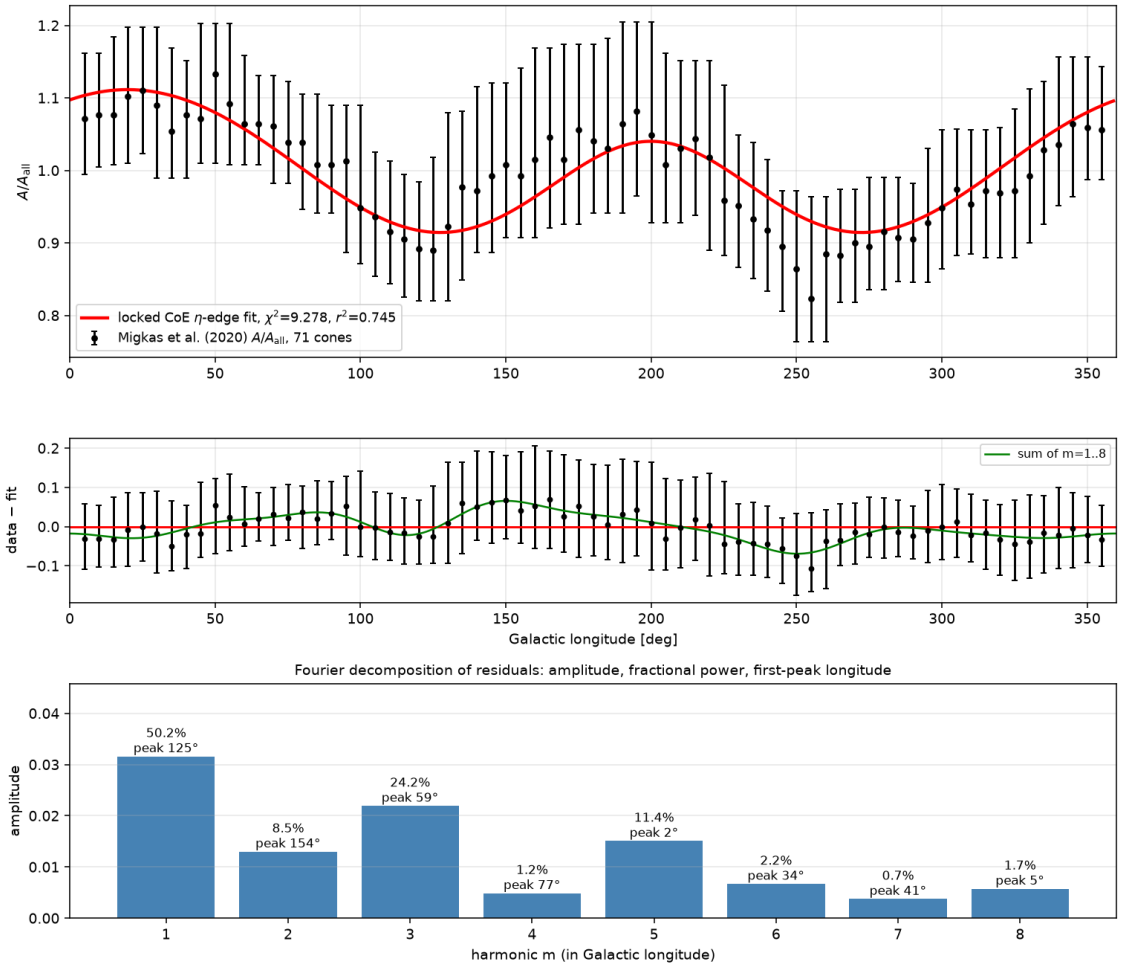


Figure 12: Top: $A/A_{\rm all}$ and locked CoE fit. Middle: data minus fit, with the $m = 1$–8 harmonic sum. Bottom: harmonic amplitudes with fractional power and first-peak longitude. Script `A_fit_residual_fourier.py`.

Residual power vs Planck, dipole removed (Fig. 13; script `A_residual_power_vs_planck_noDipole.py`). The $\ell = 1$ CMB dipole is treated as kinematic (our peculiar motion) and dropped, and $m = 1$ is dropped from the residuals, so both are renormalised over 2–8. Planck 2018 TT (`COM_PowerSpect_CMB-TT-full_R3.01.txt`, $C_\ell = 2\pi D_\ell/\ell(\ell+1)$). Shares, residual m vs Planck ℓ: 2: 17.0 vs 15.5%; 3: 48.5 vs 32.2%; 4: 2.4 vs 14.3%; 5: 22.9 vs 20.6%; 6: 4.5 vs 5.5%; 7: 1.4 vs 8.5%; 8: 3.3 vs 3.5%. *Azimuthal vs spherical harmonics.* The residual modes are azimuthal harmonics m: a 1D Fourier series $\sum_m a_m \cos[m(l - l_m)]$ in Galactic longitude around one ring of cones. Planck's ℓ are spherical multipoles: the full-sky field expanded in $Y_{\ell m}(\theta, \phi)$, with $C_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2$. A spherical multipole ℓ contributes to every ring

harmonic $|m| \leq \ell$, and a ring harmonic m receives power from every $\ell \geq |m|$, so m and ℓ are not the same quantity and the comparison is a shape comparison, not a one-to-one identification.

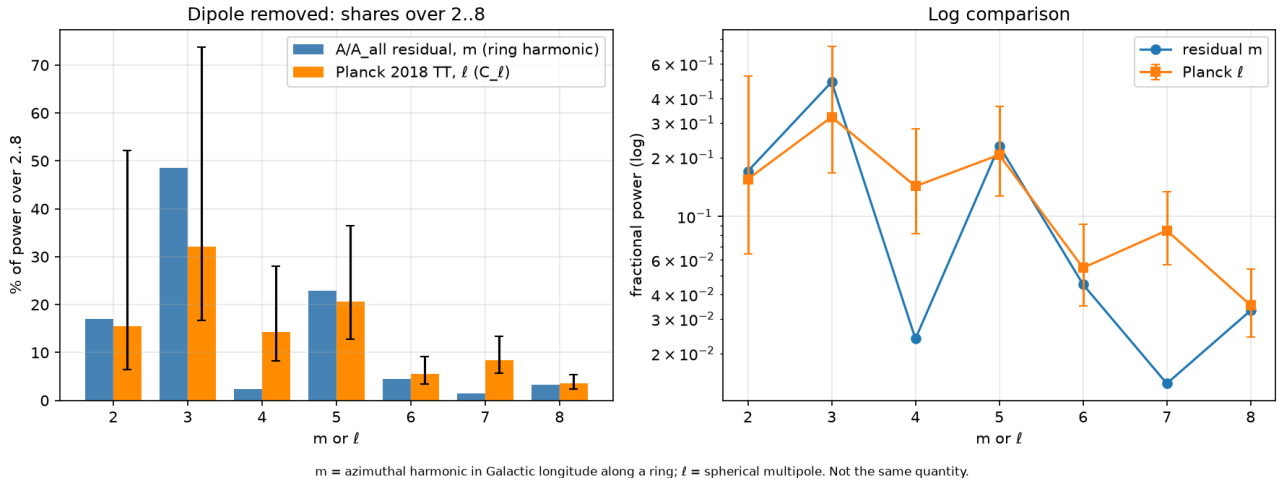


Figure 13: Fractional power over 2–8 with the dipole removed: $A/A_{\rm all}$ residual ring harmonics m (blue) and Planck 2018 TT C_ℓ (orange, 68% errors). Left linear, right log. Script `A_residual_power_vs_planck_noDipole.py`.

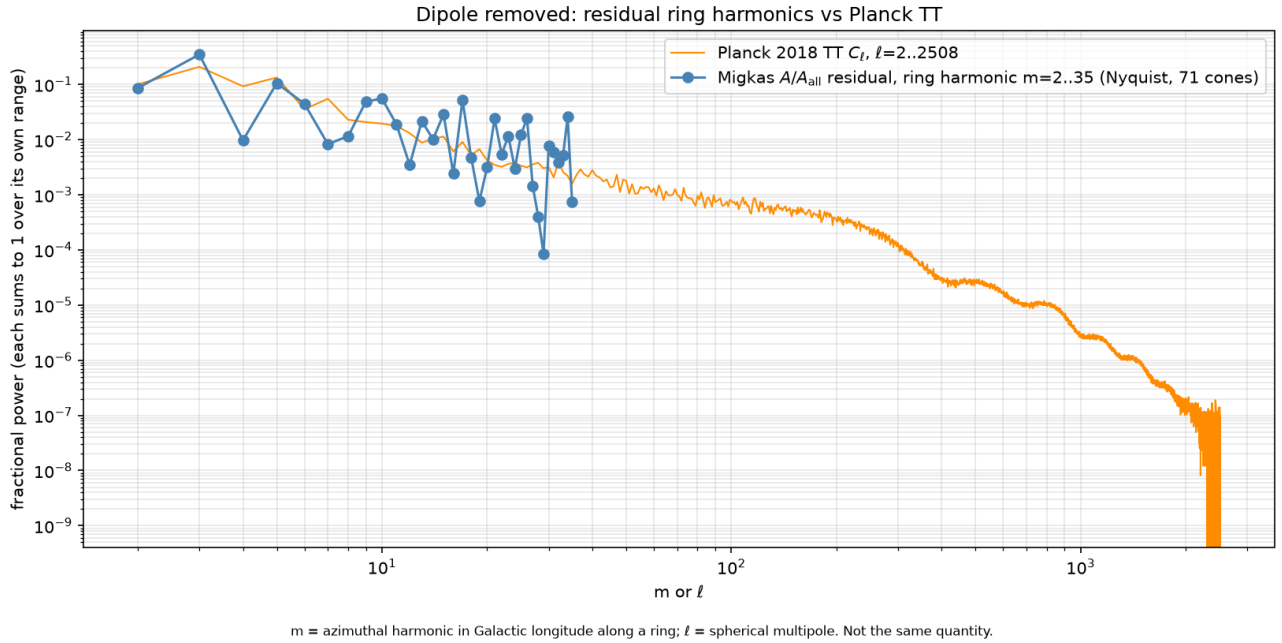


Figure 14: Dipole removed: $A/A_{\rm all}$ residual ring harmonics m = 2–35 (Nyquist limit for 71 cones at 5°) vs Planck 2018 TT C_ℓ, ℓ = 2–2508; each normalised to sum 1 over its own range. Script `A_residual_vs_planck_full.py`.

Fitting Migkas with its own Fourier series. Instead of using our simple CoE geometry, we can fit the Migkas $A/A_{\rm all}$ curve as a smooth function of longitude using its own Fourier series, $A/A_{\rm all}(l) = a_0 + \sum_{m=1}^{M}[a_m \cos ml + b_m \sin ml]$, weighted by $1/\sigma^2$. A dipole plus quadrupole ($M = 2$) already gives $R^2 = 0.831$; $M = 3$ gives 0.930. Its ring-harmonic power is then compared with Planck 2018 TT, with

the dipole kept and removed. Here m is an azimuthal ring harmonic and ℓ a spherical multipole; they are not the same quantity.

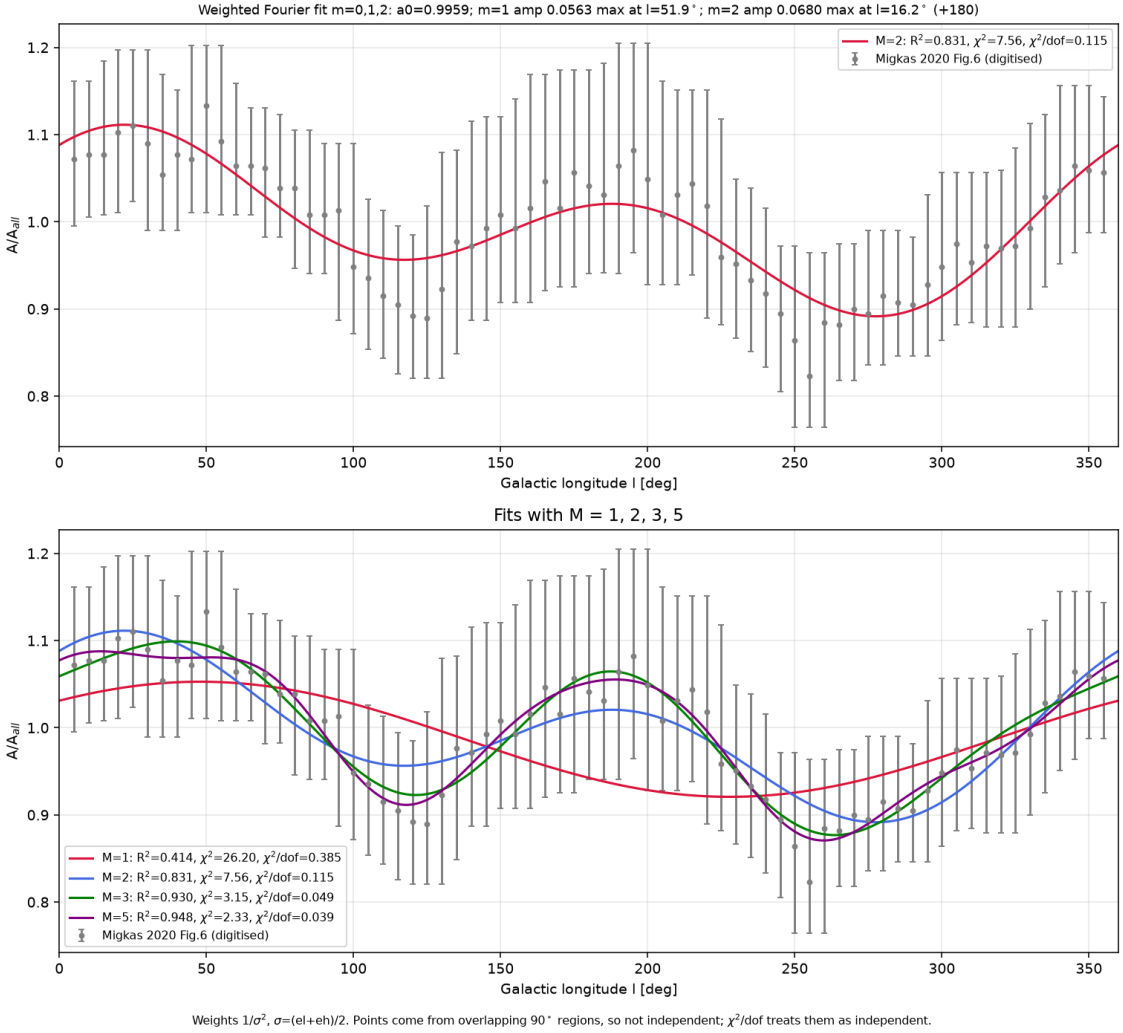


Figure 15: Weighted Fourier fits to $A/A_{\rm all}$: $M = 2$ (top) and $M = 1, 2, 3, 5$ (bottom). Script `A_fourier_fits_M1235.py`.

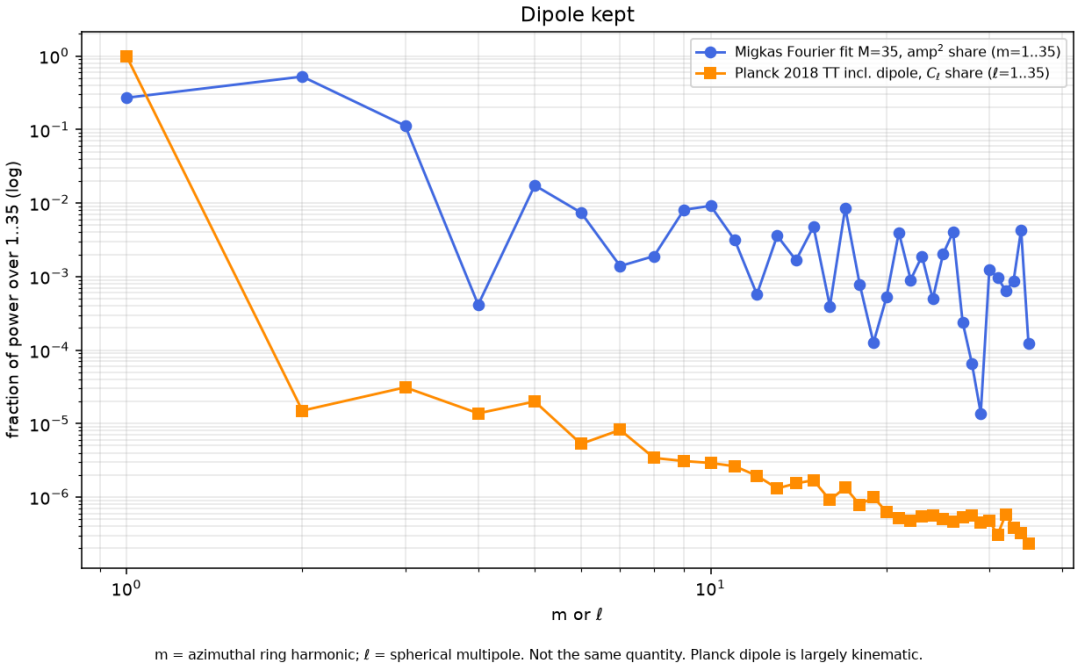


Figure 16: Dipole kept: fractional power over $m, \ell = 1$–35, Migkas Fourier fit ($M = 35$) vs Planck 2018 TT. Script `A_fourier_M35_withDipole_fig.py`.

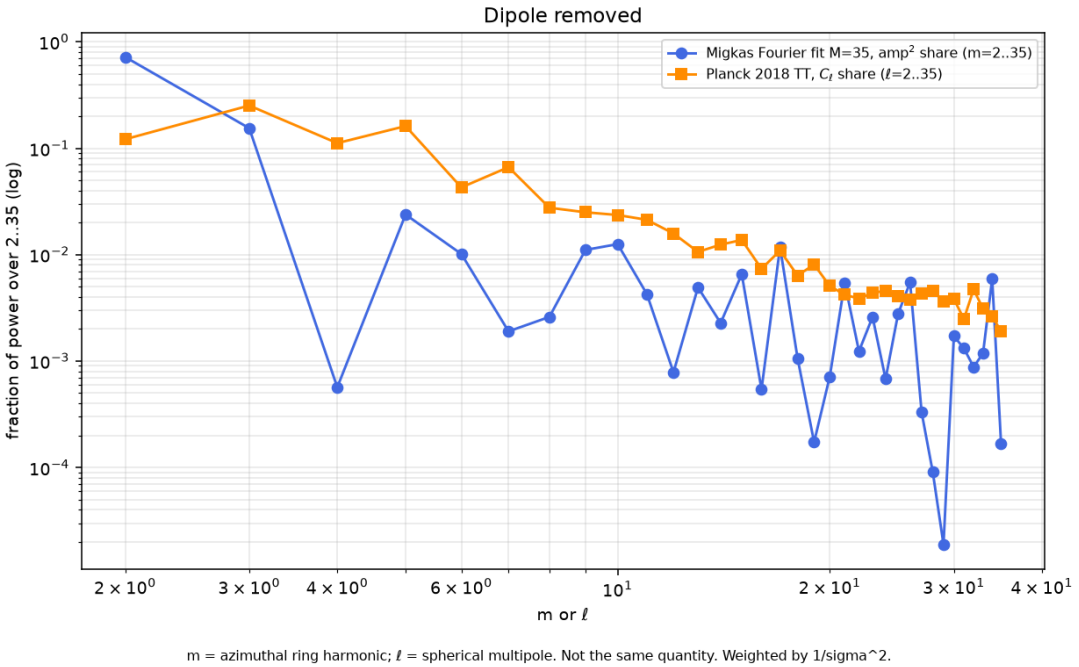


Figure 17: Dipole removed: fractional power over $m, \ell = 2$–35. Script `A_fourier_M35_noDipole_fig.py`.